

Department of Computer Science and Engineering (CSE) BRAC University

Lecture 7

CSE250 - Circuits and Electronics

MESH ANALYSIS



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Mesh Analysis

- *Mesh analysis* provides another general procedure for analysing circuits, using mesh currents as the circuit variables. Mesh analysis applies KVL to find unknown currents in a given circuit.
- A *mesh* is a loop that does not contain any other loops within it.
- *Mesh analysis is not quite as general as nodal analysis because it is only applicable to a circuit that is planar. Nonplanar circuits cannot be handled with mesh analysis.*
- *A nonplanar circuit is one that has branches that cross each other and cannot be redrawn without doing so.*

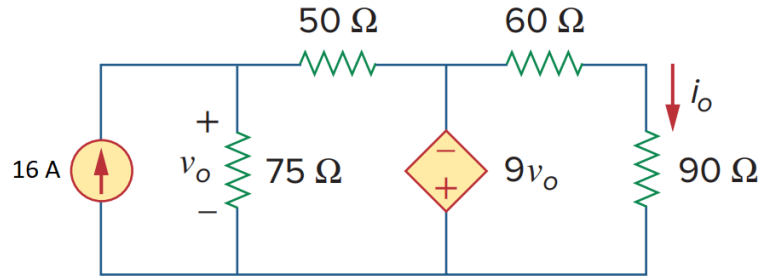
Steps to Determine Mesh Currents:

1. Assign mesh currents $i_1, i_2, \dots, i_n$ to the n meshes.
2. Apply KVL to each of the n meshes. Use Ohm's law to express the voltages in terms of the mesh currents.
3. Solve the resulting n simultaneous equations to get the mesh currents.



Planer vs Non Planer Circuit

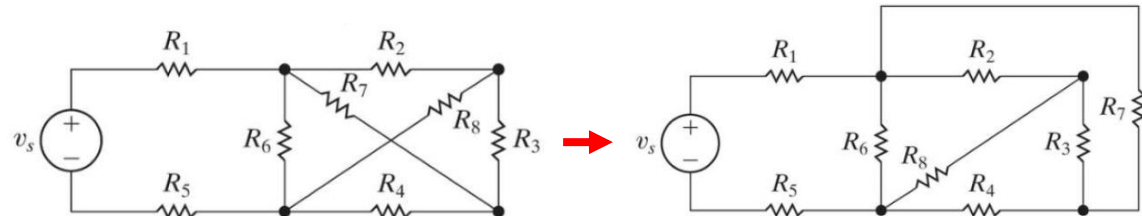
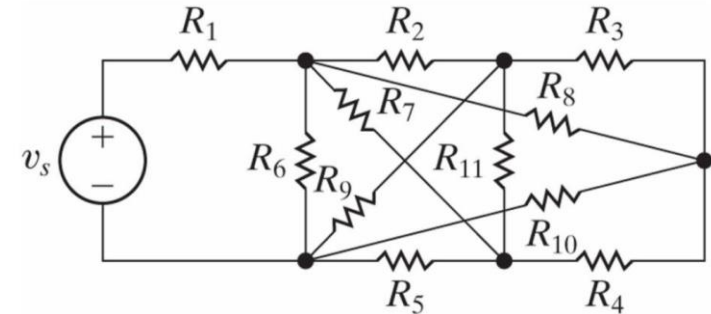
A **planar** circuit is a circuit that can be drawn on a flat surface without any wires crossing each other.



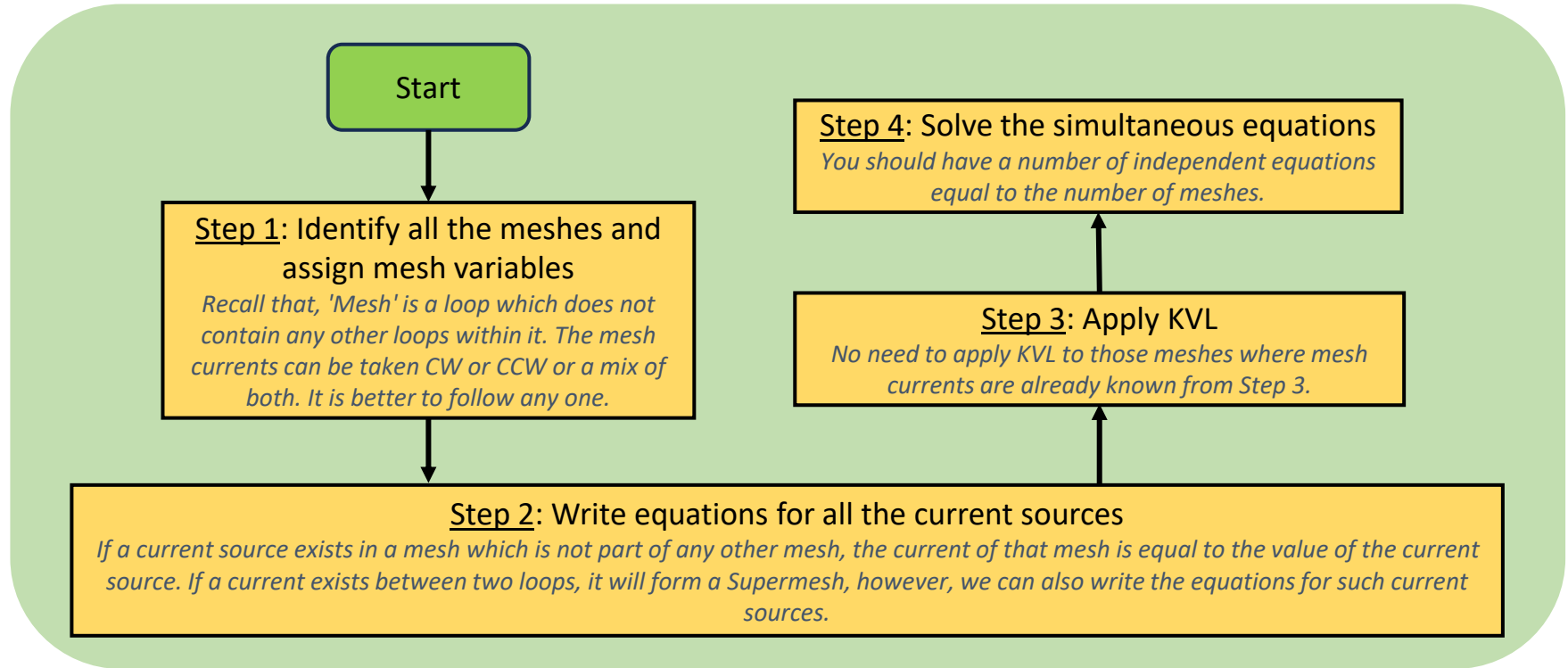
How to Identify Planar and Non-Planar Circuits?

- If the circuit can be redrawn without any wires crossing each other, then it is planar.
- If the circuit cannot be redrawn without any wires crossing each other, then it is non-planar.

A **non-planar** circuit is a circuit that cannot be drawn on a flat surface without any wires crossing each other.

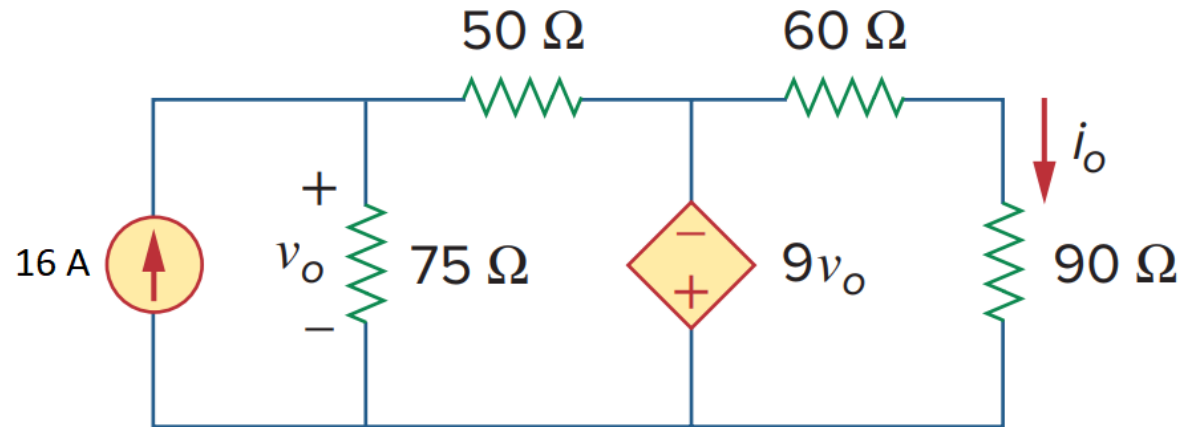


Mesh Analysis: steps



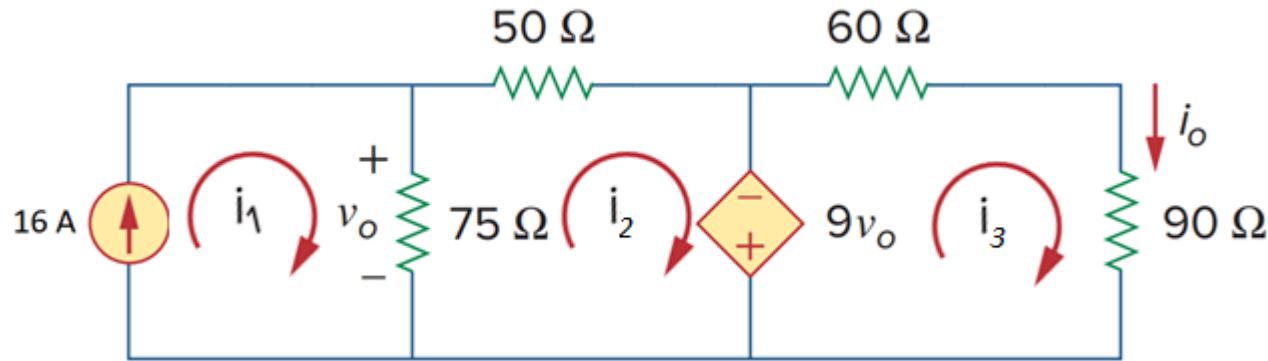
Example 1 - 1/7

Use mesh analysis, determine v_o . What is the current supplied by the dependent voltage source? What is the power of it? Is it absorbing or supplying?



Before solving the circuit using mesh analysis, recall that, "*For passive elements, current enters through the positive terminal of the voltage drop across it.*" This is according to the *passive sign convention*, current must always flow from a higher potential to a lower potential through a passive element that is absorbing power.

Example 1 - 2/7

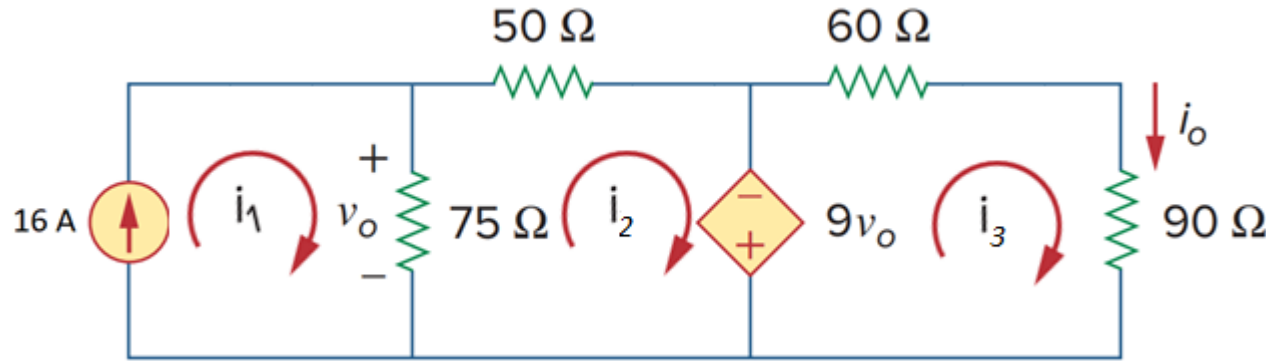


👉 First identify all the meshes (independent loops) in this circuit.

There are 3 meshes as identified in the circuit.

👉 Assign mesh currents (i_1 , i_2 , and i_3) to all the meshes. The assigned currents can be clockwise, anti-clockwise, or a combination of the two.

Example 1 - 3/7



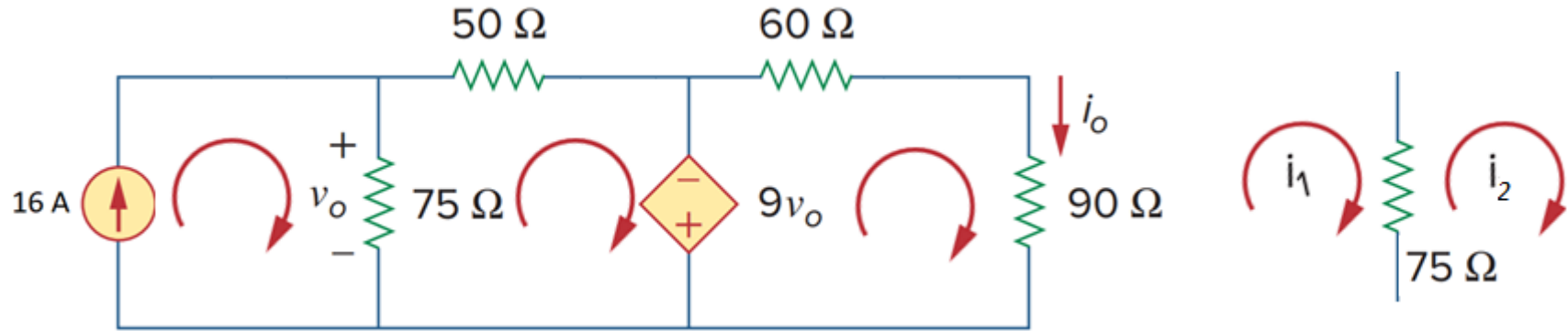
👉 The 2nd step is to apply KVL to each mesh.

Note that, we already know the mesh 1 current. i_1 and the 16 A current flow through the same wire in the same direction. We can write directly,

$$i_1 = 16 \text{ A} \text{ --- } -(i)$$

For meshes whose mesh currents are already known, we don't need to apply KVL.

Example 1 - 4/7

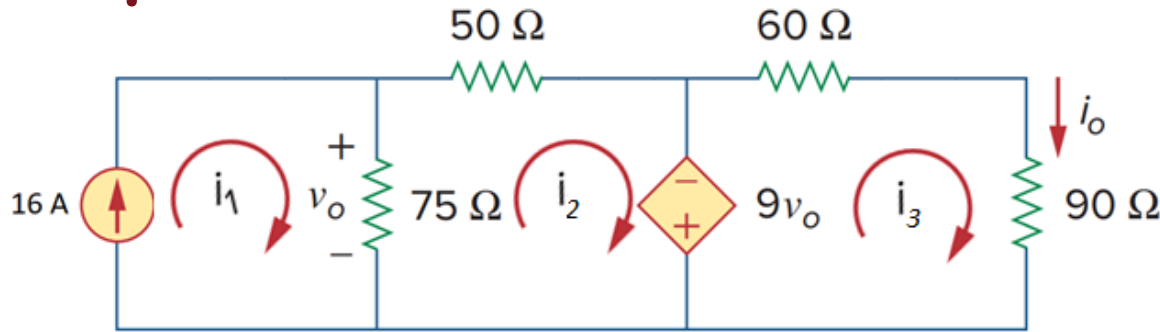


👉 Next, apply KVL to mesh 2.

$$75(i_2 - i_1) + 50i_2 - 9v_o = 0$$

Notice that, the two mesh currents (i_1 and i_2) overlap through the $75\ \Omega$. As there can be no more than a current in a wire, the resulting current through the $75\ \Omega$ will be either $i_1 - i_2$ or $i_2 - i_1$. But we won't know exactly before solving. As we are moving in the direction of i_2 , we take $i_2 - i_1$ as the resulting current and the KVL equation is written accordingly.

Example 1 - 5/7



$$75 (i_2 - i_1) + 50i_2 - 9v_0 = 0 \text{ [from the previous slide]}$$

Now we have to replace v_0 in terms of the mesh currents as the mesh equations should not contain unknowns other than the mesh currents.

v_0 is the voltage drop across the 75Ω resistor. With the polarity of v_0 given,

$$v_0 = 75 (i_1 - i_2)$$

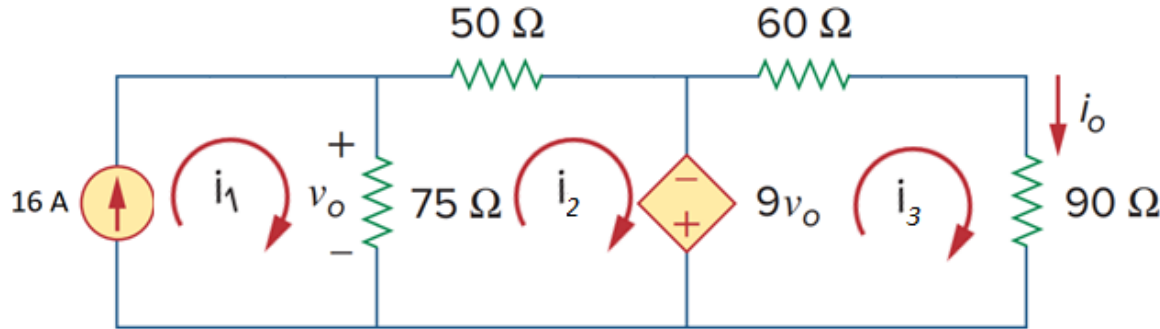
Substituting,

$$75 (i_2 - i_1) + 50i_2 - 9 \times 75 (i_1 - i_2) = 0$$

$$750 i_1 - 800 i_2 = 0 \text{ --- (ii)}$$



Example 1 - 6/7



👉 Next, apply KVL to mesh 3.

$$9v_o + 60i_3 + 90i_3 = 0$$

Substituting $v_o = 75(i_1 - i_2)$ for v_o ,

$$9 \times 75(i_1 - i_2) + 60i_3 + 90i_3 = 0$$

After simplifying,

$$675i_1 - 675i_2 + 150i_3 = 0 \text{ --- (iii)}$$

Example 1 - 7/7

We have derived the three mesh equations,

$$i_1 = 16 \text{ A}$$

$$750 i_1 - 800 i_2 = 0$$

$$675 i_1 - 675 i_2 - 150 i_3 = 0$$

Solving,

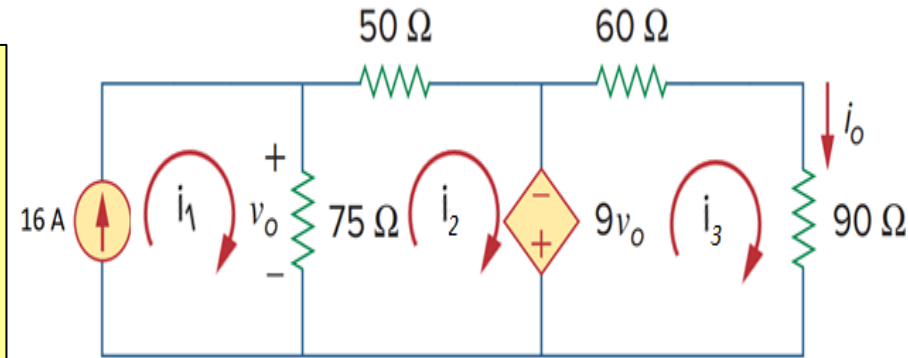
$$i_1 = 16 \text{ A}; \quad i_2 = 15 \text{ A}; \quad i_3 = -4.5 \text{ A};$$

So,

$$v_0 = 75(i_1 - i_2) = 75(16 - 15) = 75 \text{ V}$$

Current supplied (entering into the -ve terminal) by the dependent source is,

$$i_2 - i_3 = 15 - (-4.5) = 19.5 \text{ A}$$



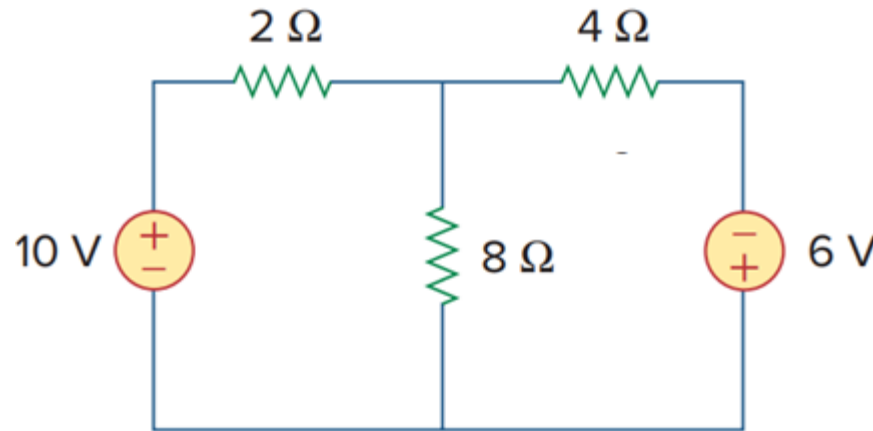
Power supplied by the dependent source is thus,

$$\begin{aligned} p &= -vi = 9v_0 \times 19.5 \\ &= 9 \times 75 \times 19.5 \\ &= 13162.5 \text{ W} \end{aligned}$$



Problem 1

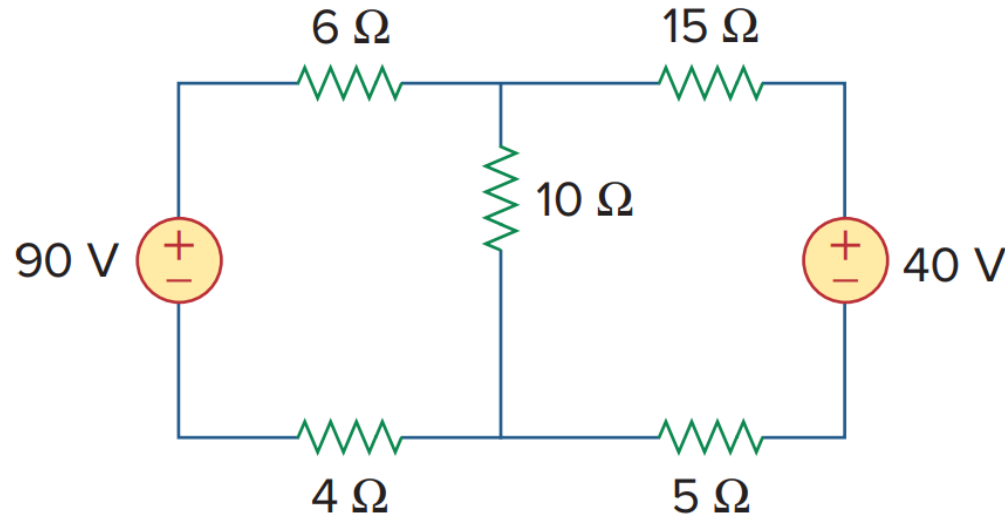
- Perform *branch current analysis* to determine the current absorbed by the 6 V source in the following circuit.
- Perform *mesh analysis* to determine the current absorbed by the 6 V source in the following circuit.



Ans: -2.5 A

Problem 2

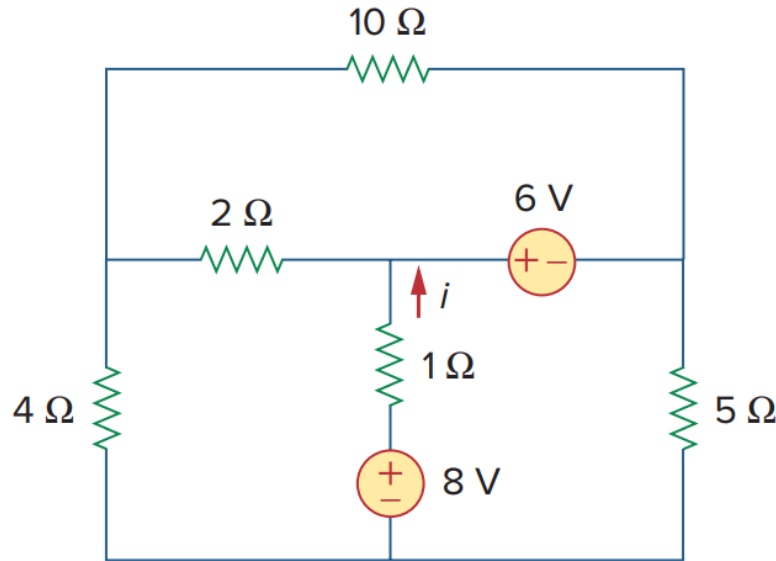
- Calculate the current through the $10\ \Omega$ resistor using mesh analysis.



<u>Ans:</u> $I_{10\Omega} = 4.4\text{ A}$

Problem 3

- Calculate the current i using mesh analysis.

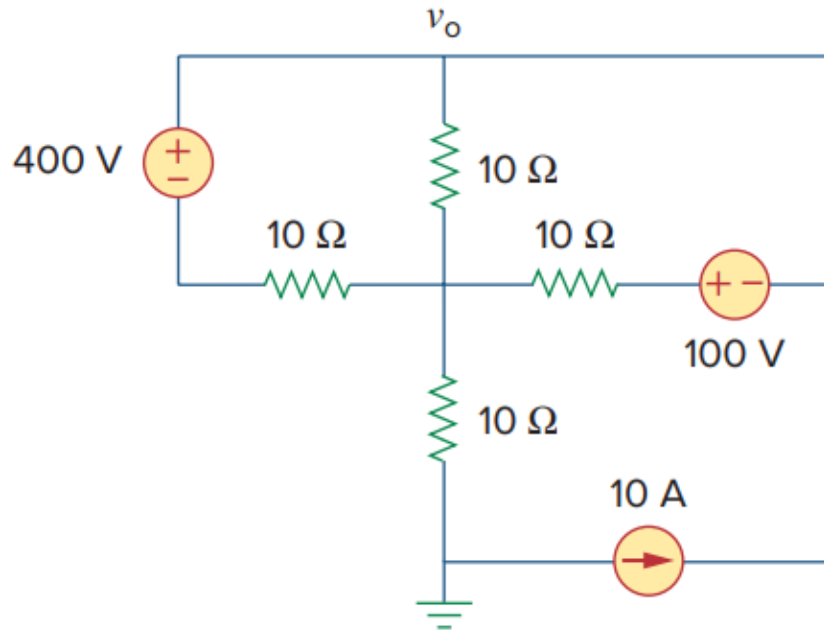


Ans: $i = 1.188\text{ A}$



Problem 4

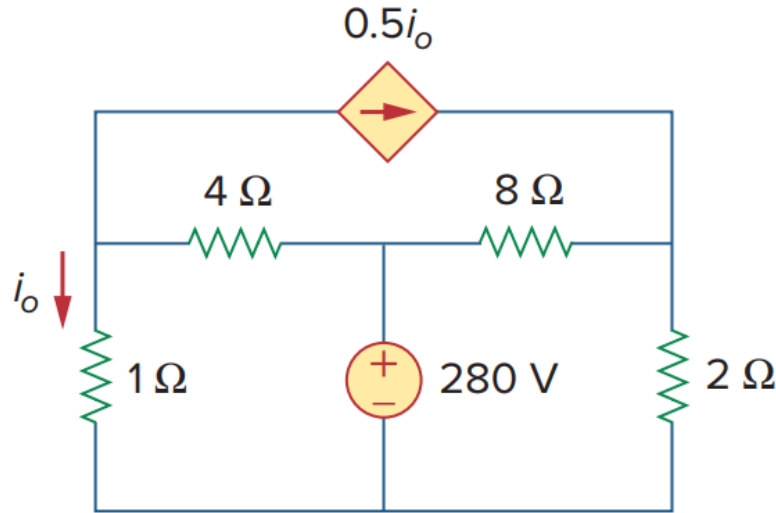
- Apply mesh analysis to find v_o in the following circuit.



Ans: $v_o = 233.3 \text{ V}$

Problem 5

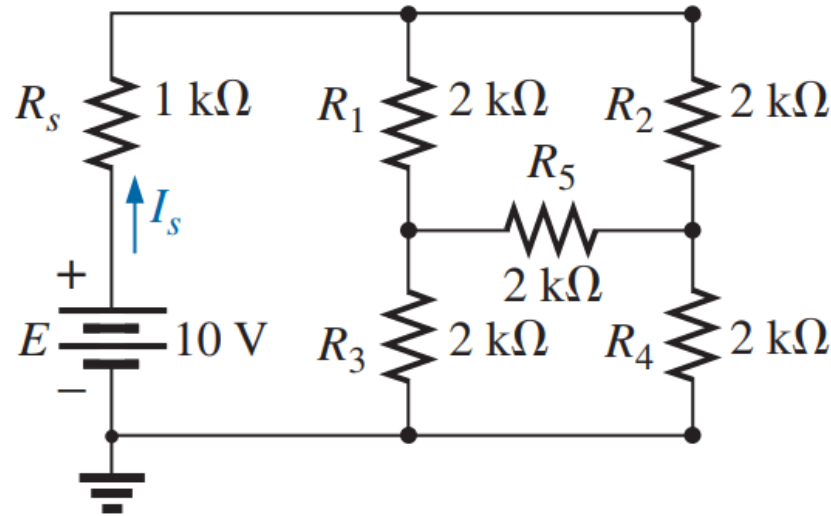
- Find i_0 using mesh analysis. What is the voltage across the $0.5i_0$ source?



Ans: $i_0 = 40 \text{ A}; \pm 48 \text{ V}$

Problem 6

- Determine the current through the source resistor R_s using mesh analysis.



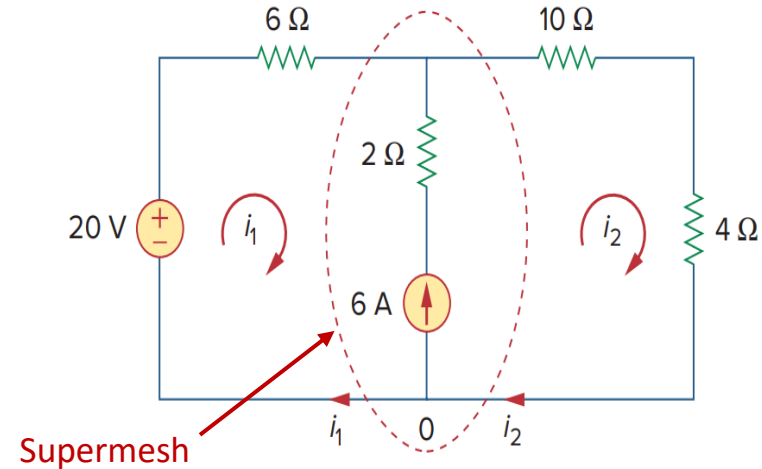
Ans: $i_s = 3.33 \text{ mA}$

Analysis with current source betⁿ loops

■ **CASE 1** When a current source (dependent or independent) exists only in one mesh, we simply set the current at that mesh equal to the current of the current source. (We have already seen this in [example 1](#)).

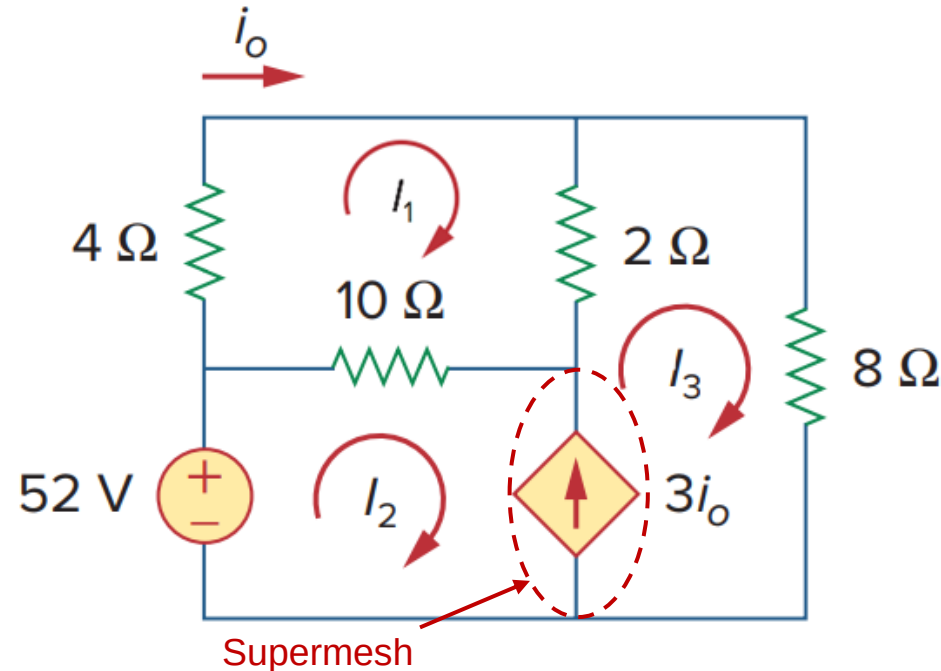
■ **CASE 2** When a current source (dependent or independent) exists between two meshes, the two meshes form a generalized mesh or supermesh.

In other words, a *supermesh* results when two meshes have a (dependent or independent) current source in common.



Example 2 - 1/5

- Find i_o using mesh analysis. Also, calculate the voltage across the $3i_o$ source.



Step 1: Identify all the meshes and assign mesh variables to each of the meshes.

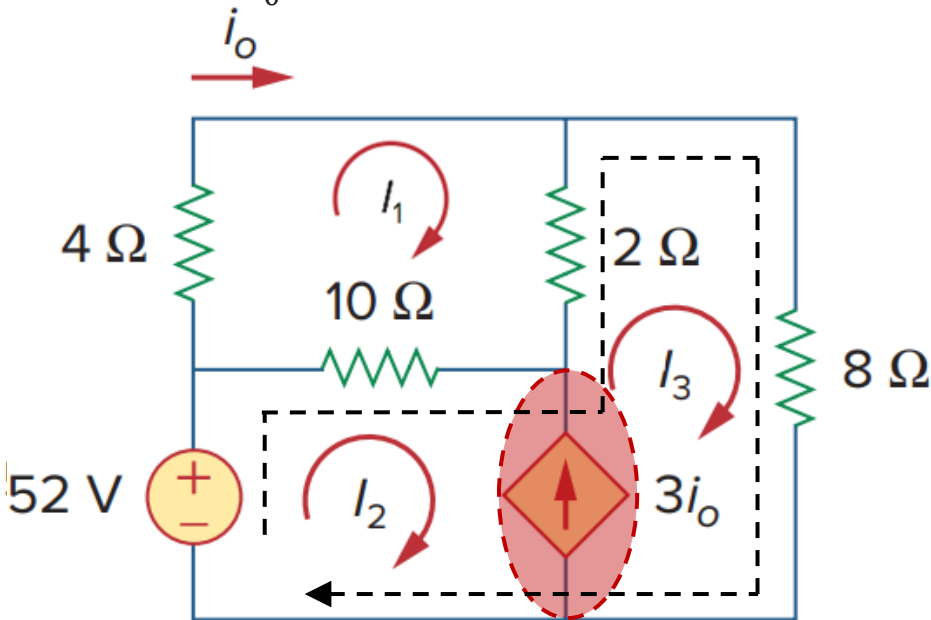
Check for supermeshes. Check if a current source (dependent or independent) is connected between two meshes. There can be multiple supermeshes in a circuit.

In this circuit, the $3i_o$ current source forms a supermesh between meshes 2 and 3.

We need to handle such conditions differently because there is no way to know the voltage across a current source in advance.

Example 2 - 2/5

- Find i_0 using mesh analysis. Also, calculate the voltage across the $3i_0$ source.



Step 2: Apply KVL to each of the meshes.

KVL to the mesh 1,

$$4i_1 + 2(i_1 - i_3) + 10(i_1 - i_2) = 0$$

$$\Rightarrow 16i_1 - 10i_2 - 2i_3 = 0 \text{ --- (i)}$$

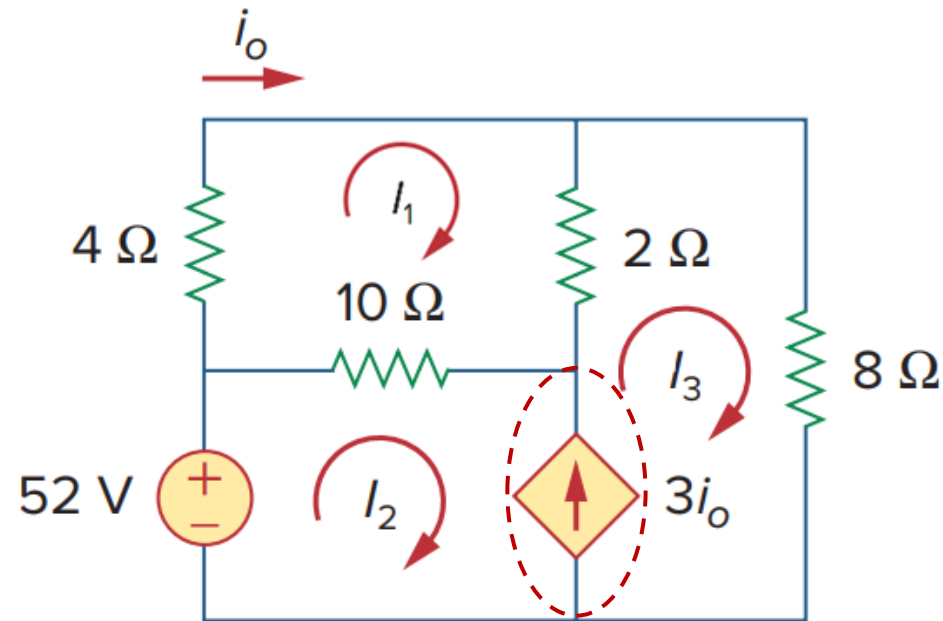
Next, ignore the current source that forms the supermesh and apply KVL to the corresponding meshes together. Careful with the current notations. Applying KVL to the supermesh along the black dotted line shown in the figure,

$$-52 + 10(i_2 - i_1) + 2(i_3 - i_1) + 8i_3 = 0$$

$$\Rightarrow 12i_1 - 10i_2 - 10i_3 = -52 \text{ --- (ii)}$$

Example 2 - 3/5

- Find i_0 using mesh analysis. Also, calculate the voltage across the $3i_0$ source.



We have 2 equations, 3 variables, and no remaining mesh for KVL.

The 3rd equation required, can be found by applying KCL to the supernode.

$$i_3 - i_2 = 3i_0$$

Now replace i_0 in terms of the mesh currents. It can be seen from the figure that, $i_0 = i_1$. Substituting,

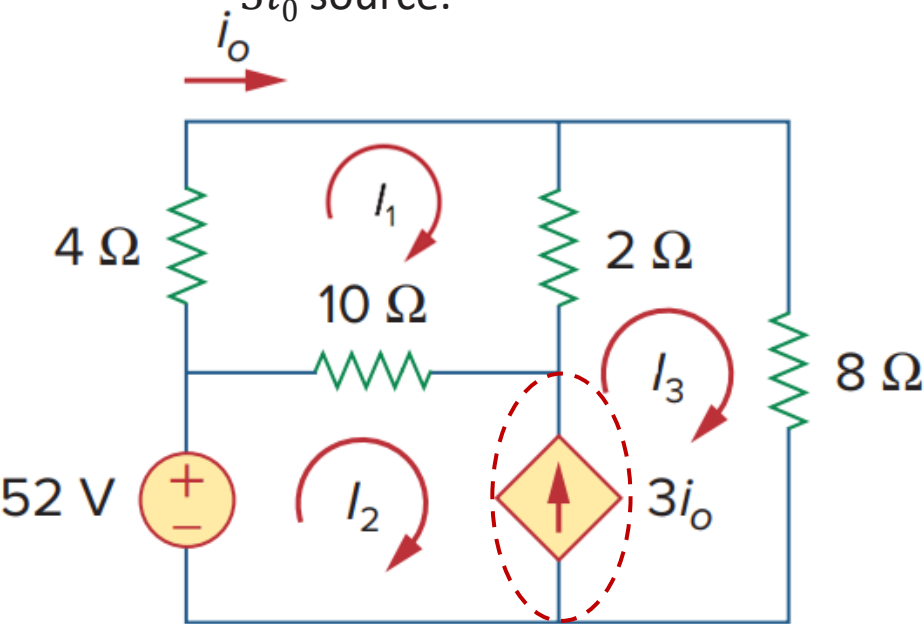
$$i_3 - i_2 = 3i_1$$

$$\Rightarrow 3i_1 + i_2 - i_3 = 0 \text{ --- (iii)}$$



Example 2 - 4/5

- Find i_0 using mesh analysis. Also, calculate the voltage across the $3i_0$ source.



We have derived the three equations,

$$16i_1 - 10i_2 - 2i_3 = 0$$

$$12i_1 - 10i_2 - 10i_3 = -52$$

$$3i_1 + i_2 - i_3 = 0$$

Solving,

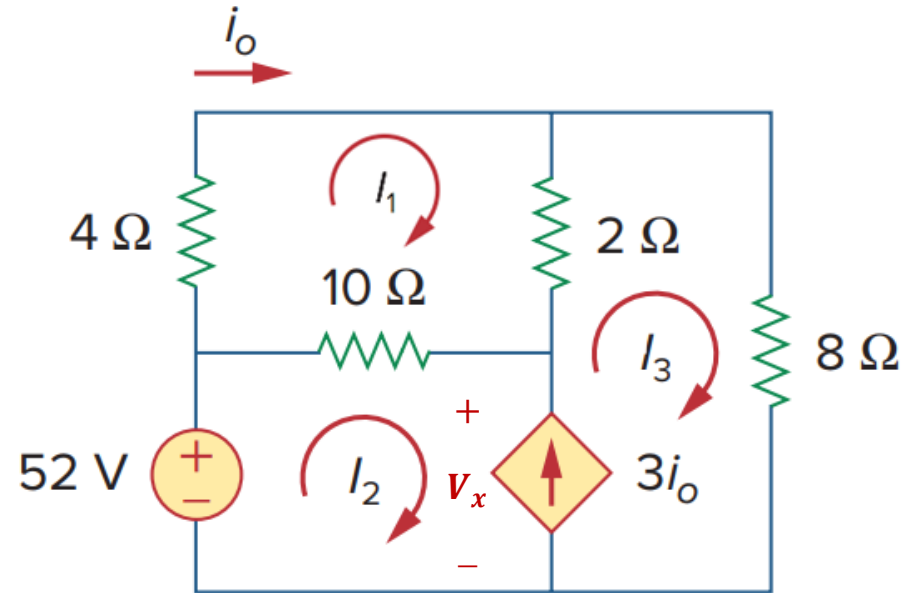
$$i_1 = 1.5 \text{ A}; \quad i_2 = 1.25 \text{ A}; \quad i_3 = 5.75 \text{ A}$$

So, $i_0 = i_1 = 1.5 \text{ A}$

To calculate the voltage across the $3i_0$ dependent source, we have to apply KVL to either loop 2 or loop 3.

Example 2 - 5/5

- Find i_o using mesh analysis. Also, calculate the voltage across the $3i_o$ source.



Let the voltage across the $3i_o$ source is V_x as indicated in the figure.

Applying KVL to the loop 2,

$$-52 + 10(i_2 - i_1) + V_x = 0$$

$$\Rightarrow V_x = 52 - 10(1.25 - 1.5) = 54.5 \text{ V}$$

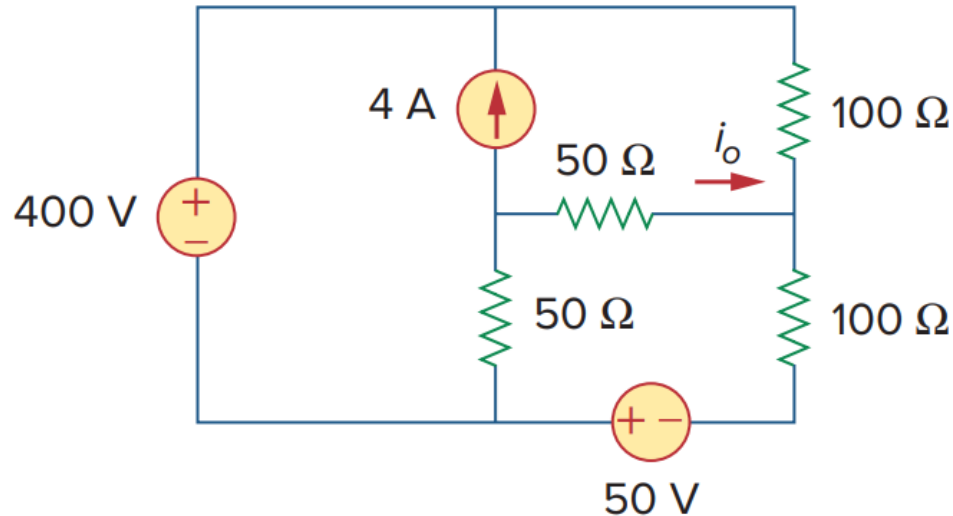
As observed by the polarities of voltage and current, the dependent source is supplying power.

$$p = +vi = 54.5 \times 3i_o = 54.5 \times 3i_1$$

$$\Rightarrow p = 54.5 \times 3 \times 1.5 = 245.25 \text{ W}$$

Problem 7

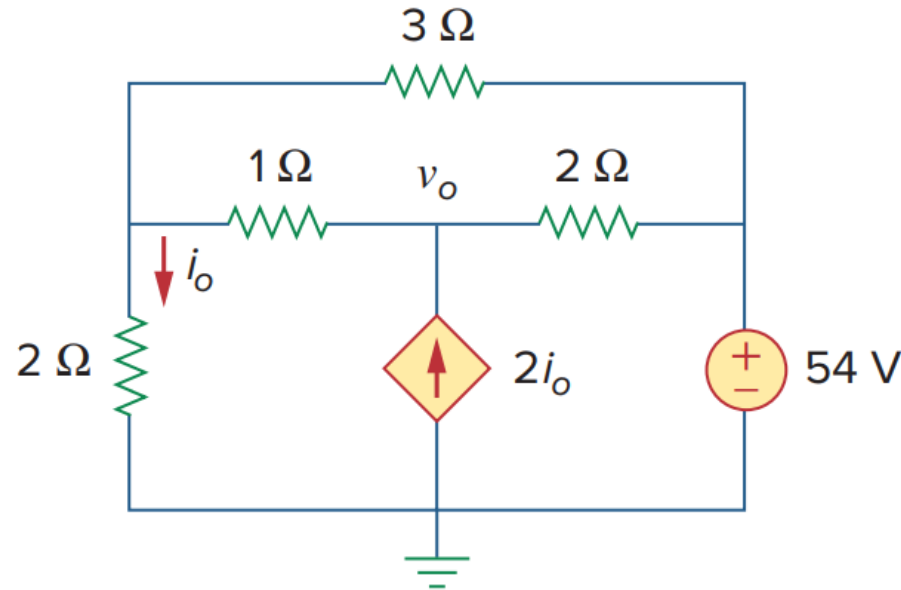
- Find i_0 using mesh analysis.



Ans: $i_0 = -2.5 \text{ A}$

Problem 8

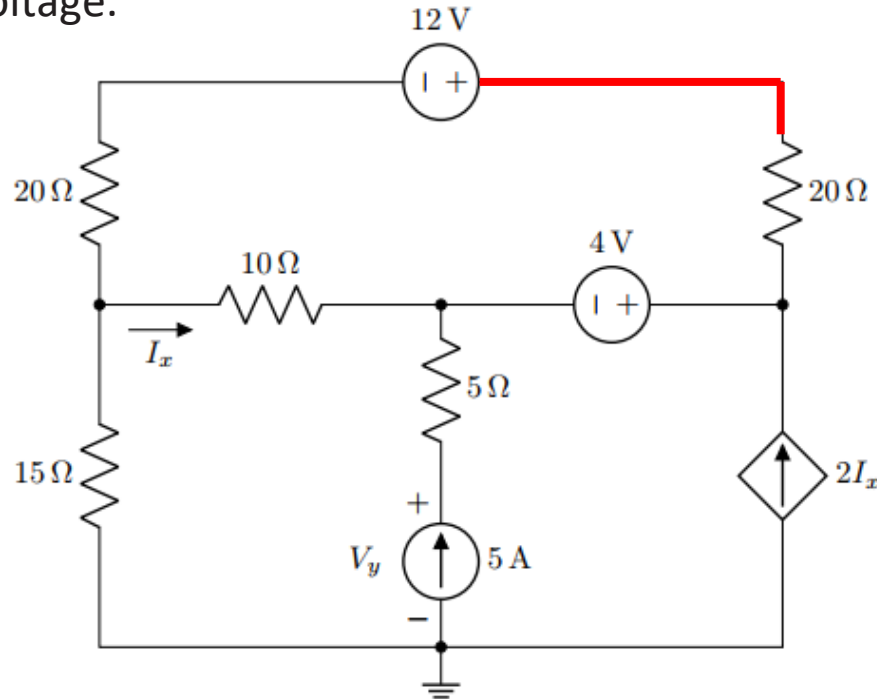
- Find i_0 using mesh analysis. Determine the node voltage v_0 .



Ans: $i_0 = 36 \text{ A}$; $v_0 = 114 \text{ V}$

Problem 9

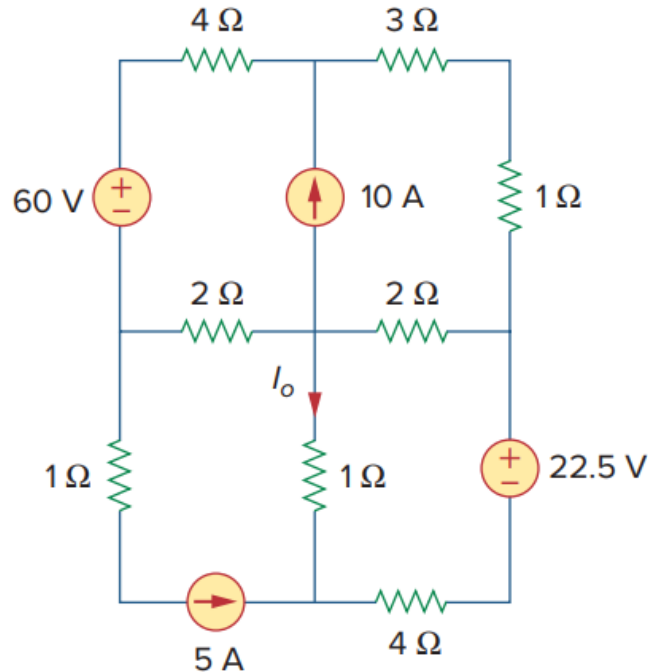
- Use Mesh Analysis to analyze the circuit. Find V_y . Determine the red colored node voltage.



Ans: $V_y = 68\text{ V}$; $V_{red} = 43\text{ V}$

Problem 10

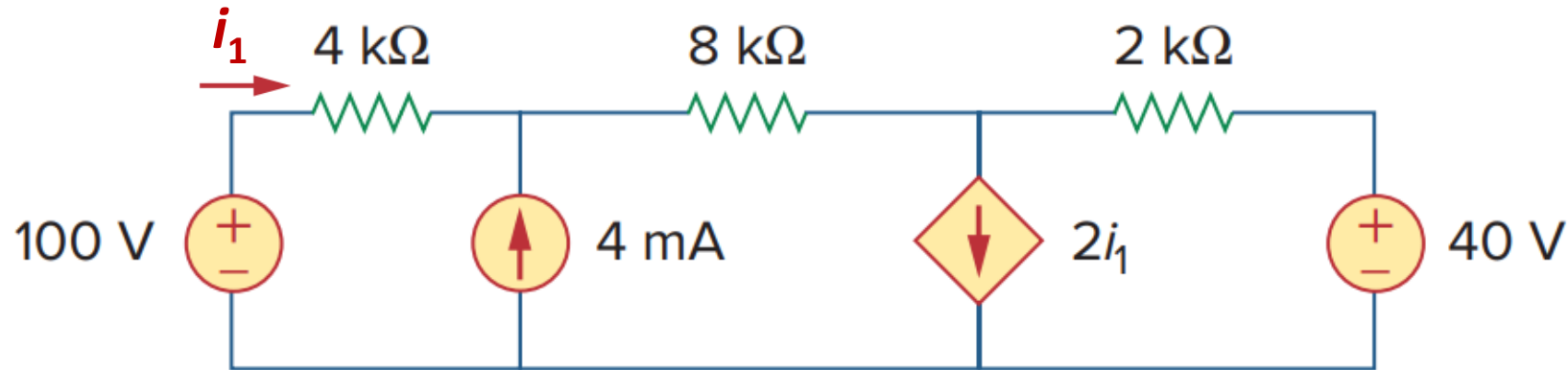
- Derive the mesh equations for the following circuit. Determine i_0 .



Ans: $i_0 = -3.62 \text{ A}$

Problem 11

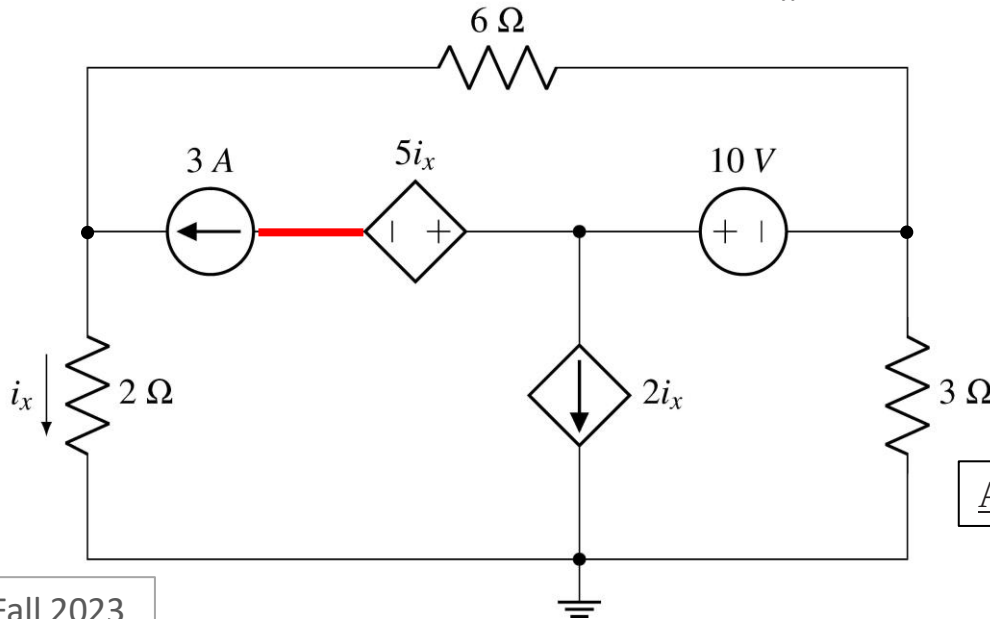
- Find the mesh currents.



Ans: $\pm 2 \text{ mA}$; $\pm 6 \text{ mA}$; $\pm 2 \text{ mA}$

Problem 12

- (i) Use nodal analysis to find i_x . Determine the voltage across the 3 A source.
- (ii) Use mesh analysis to find i_x . Determine the voltage of the red colored node.



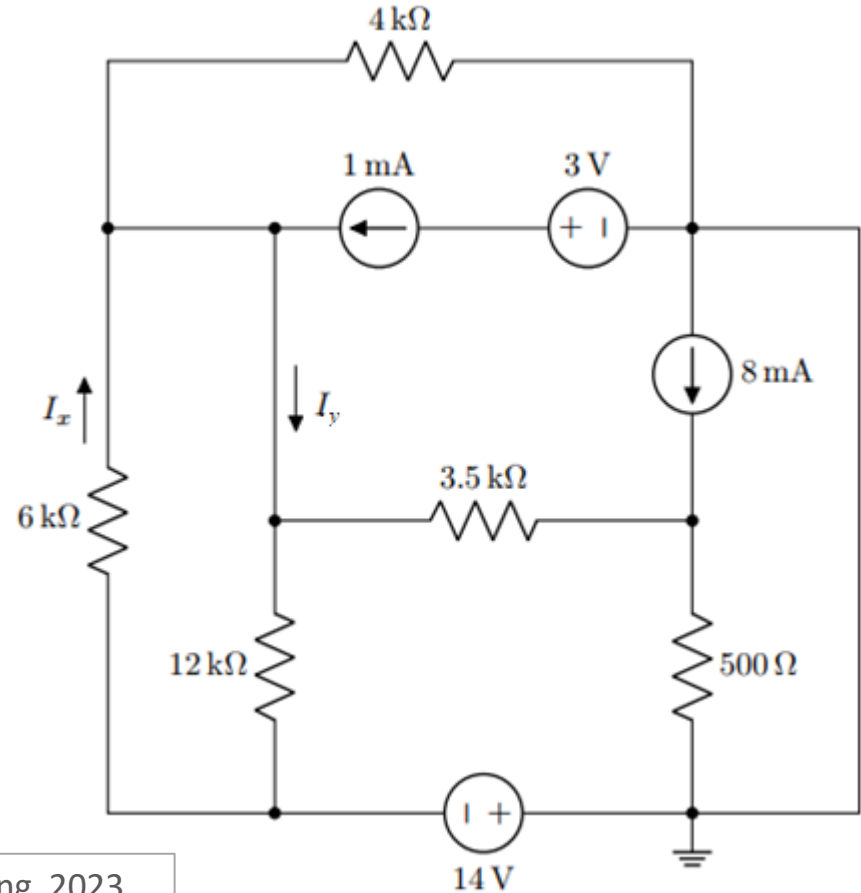
Ans: $i_x = 1.059 \text{ A}$; $v_{3A} = \pm 6.94 \text{ V}$; $v_{\text{red}} = -4.82 \text{ V}$

Fall 2023



Problem 13

- Use mesh analysis to analyze the circuit. Find I_x .
- Determine the current I_y .
- Now repeat using Nodal analysis.
- Compare the two methods in solving this circuit.

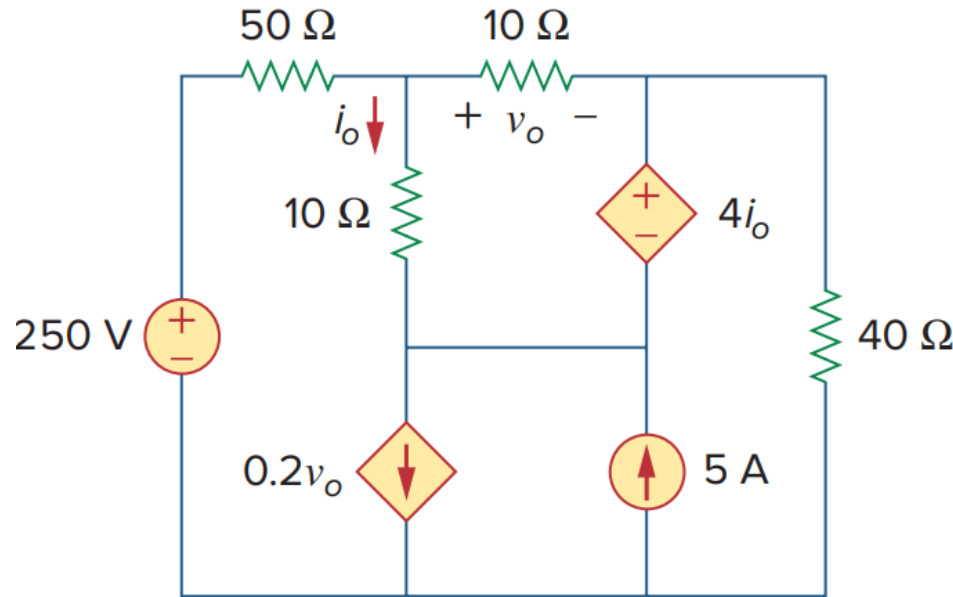


Ans: $I_x = -2 \text{ mA}$; $I_y = -0.5 \text{ mA}$

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Problem 14

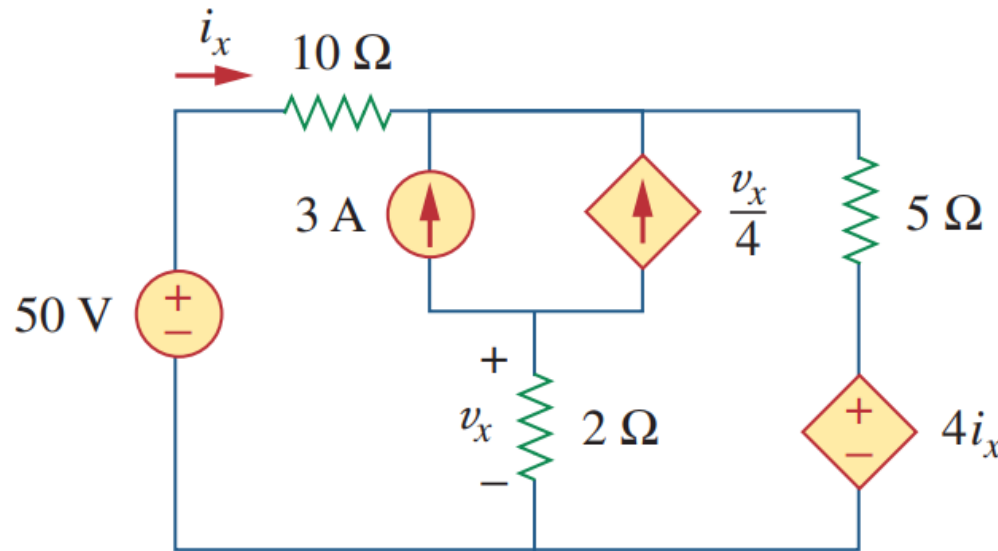
- Use mesh analysis to determine v_o and i_o . What is the voltage across the 5 A source?



Ans: $v_o = 2.941 \text{ V}$; $i_o = 0.49 \text{ A}$

Problem 15

- Use mesh analysis to determine v_x and i_x . What is the voltage across the 3 A source?



Ans: $v_x = -4\text{ V}$; $i_x = 2.105\text{ A}$

Nodal vs Mesh Analysis

- Given a network to be analysed, how do we know which method is better or more efficient?
The choice of the better method is dictated by two factors:

■ Nature of the network

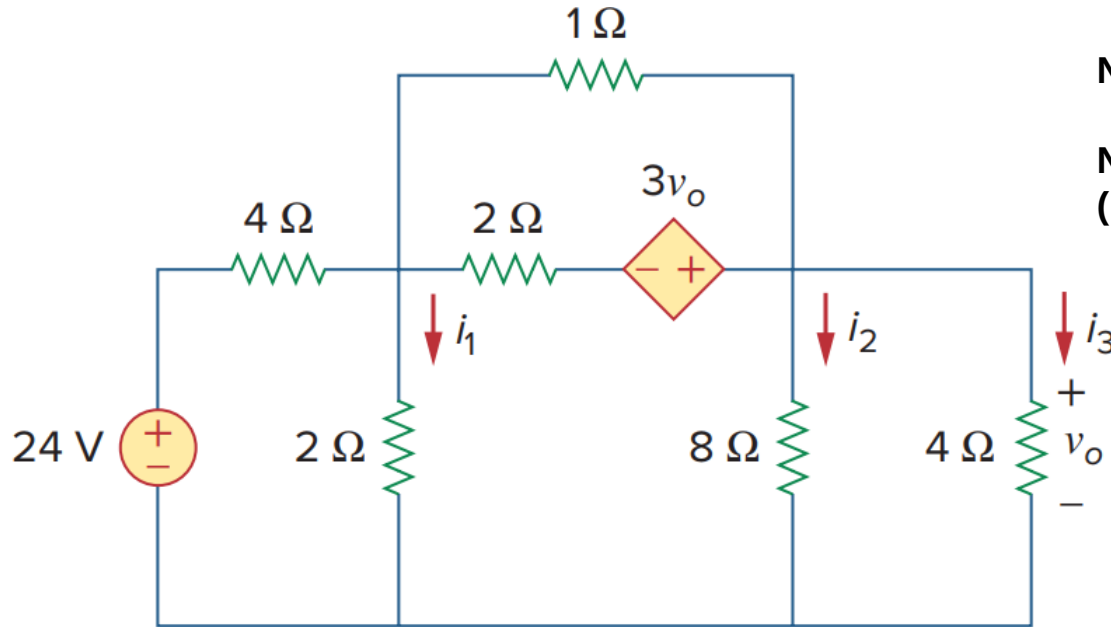
Mesh analysis is easier for networks that contain many series-connected elements, voltage sources, or supermeshes	Nodal analysis is easier for networks with parallel connected elements, current sources, or supernodes.
A circuit with fewer nodes than meshes is better analysed using nodal analysis, and A circuit with fewer meshes than nodes is better analysed using mesh analysis. The key is to select the method that results in the smaller number of equations.	

■ Information required

Mesh analysis is easier if branch or mesh currents are required	Nodal analysis is easier if node voltages are required
As we shall see in CSE251, mesh analysis is the only method to use in analysing transistor circuits. But mesh analysis cannot easily be used to solve an op amp circuit, because there is no direct way to obtain the voltage across the op amp itself. For nonplanar networks, nodal analysis is the only option.	

Problem 16

- Which method, nodal or mesh, is more convenient for solving the circuit? Derive the equations that correspond to the convenient one.



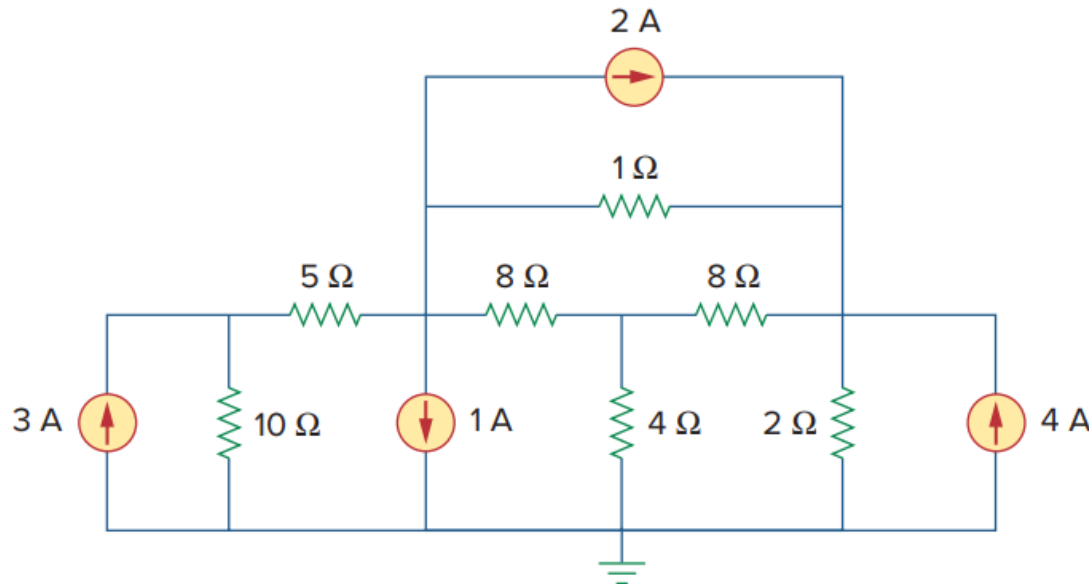
Number of mesh = 4

Number of unknown node variables = 2
(If ground is placed at the bottom node)

So, USE Nodal Analysis

Problem 17

- Count how many nodes and meshes there are in this circuit. What is the bare minimum of variables that need to be considered for both nodal and mesh analysis? Which of these methods is the most convenient for solving the circuit? Determine the equations that correspond to the convenient one.

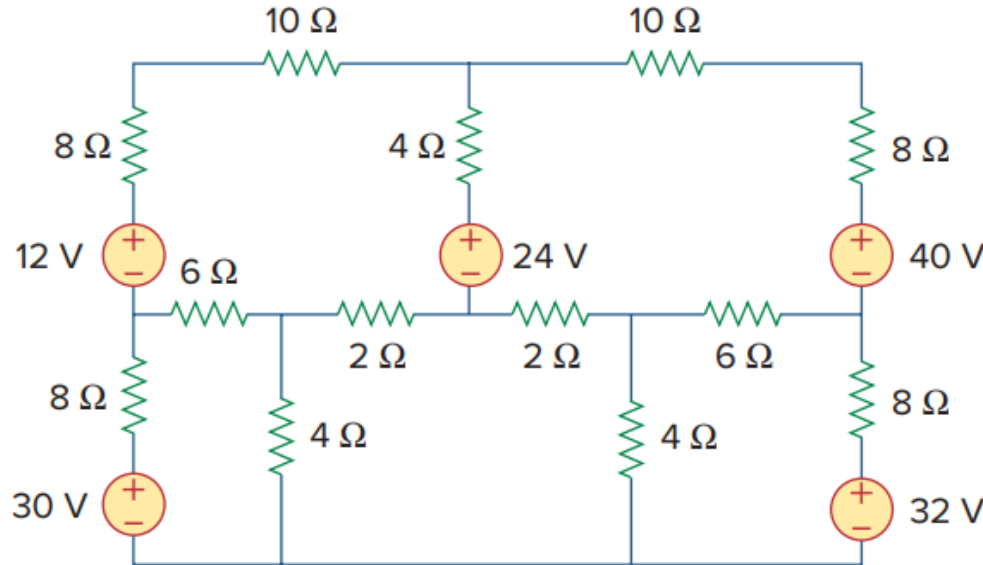


Ans:

- # of nodes = 4;
- # of meshes = 7;
- minimum # of variables for nodal analysis = 4;
- minimum # of variables for mesh analysis = 7.

Problem 18

- Count how many nodes and meshes there are in this circuit. What is the bare minimum of variables that need to be considered for both nodal and mesh analysis? Which of these methods is the most convenient for solving the circuit? Determine the equations that correspond to the convenient one.



Ans:

- # of nodes = 14;
- # of meshes = 5;
- minimum # of variables for nodal analysis = 6;
- minimum # of variables for mesh analysis = 5.

Practice Problems

- Additional recommended practice problems: [here](#)
- Other suggested problems from the textbook: [here](#)

Thank you for your attention

Course Outline: broad themes

